Sasha Lopoukhine

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Education

University of Cambridge Cambridge, UK
PhD in Computer Science supervised by Tobias Grosser *Jan 2023 — Present*

Developing assembly-level MLIR representations and compiler tooling for linear-algebra micro-kernels on AI accelerators.

Ongoing: Building MLIR representations and rewrite passes for x86 AVX-512 to generate L1-resident matrix multiplication kernels matching the state of the art.

CGO 2025: A Multi-level Compiler Backend for Accelerated Micro-kernels Targeting RISC-V ISA Extensions Developed an MLIR-based linear algebra micro-kernel backend that targets ETH's Snitch Core, yielding 95% FPU utilization and 94% of theoretical maximum throughput for our chosen kernels. Neural networks compiled via this micro-kernel outperform output of LLVM by >20x.

CGO 2025: Sidekick compilation with xDSL Leading development of xDSL, a Python-native, MLIR-compatible compiler framework used to prototype and ship compiler infrastructure in Python. Now at 1M+ PyPI downloads and 125+ contributors; production use includes quantum compilers (Riverlane, Xanadu, Eclipse) and Google LiteRT; research integrations include Devito and PSyclone. Featured in GitHub Maintainer Month added to the CPython performance benchmark suite; cited in PEP 747.

Supervised four BA and one MA dissertation projects, one of which won silver at the CGO 2024 student research competition.

University of Cambridge Cambridge, UK
BA in Computer Science - II.i *2012 — 2015*

Received Bob Diamond prize for Computer Science.

Employment History

Student Investment Partner (promoted from Venture Fellow) Oct 2024 — Present
Creator Fund *Cambridge, UK*

Sourced Cambridge-based startups and conducted technical due diligence for investment decisions.

Co-Founder & Chief Engineering Officer Sep 2021 — Jan 2022
Cub3 *Los Angeles, CA*

Co-founded Cub3; led engineering from zero to MVP of a Web3 fan-engagement platform.

Software Engineer Mar 2018 — Sep 2021
Apple *Cupertino, CA*

Curated Library – iOS 13 feature shipped to **1B+ devices**.
- Built the Years/Months/Days timeline from photo metadata. **Patent**: US20200356227A1 – **Curated Media Library**
- Profiled and optimized Years/Months/Days timeline computation to ensure low-latency processing when taking photos.

On-disk graph – Implemented a graph database for Memories data (~1000x prior capacity).
- Unlocked on-demand availability of all Memories, automatic song selection, and next memory suggestion.
- Led the migration of existing features with minimal regressions, reducing benchmark runtime **9×** (90 ⟶ 10 min).

Next memory suggestion – Developed a recommendation engine for Memories.

Apple Music in Memories – Led feature extraction from the Memories DB for music recommendation.
- Mentored the team on Swift and software architecture.

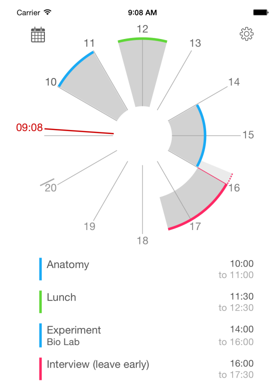
iOS Engineer — Jan 2016 — Sep 2017
Snips — *Paris, France*

Keyboard extension – system-wide iOS keyboard suggesting information from the Snips Personal Knowledge Graph.
- Integrated email, calendar, and contacts for real-time context as the user typed.
- Developed a binary-encoded language model for auto-completion and next-word prediction; loaded in milliseconds.

Snips Platform – Labs - prototyped natural language parsers and voice assistants on iOS.
- Built iOS voice assistant handling contextual queries (e.g. "Order cab to restaurant I went to last week.") via CKY parsing.
- Built query gap-filling engine for incomplete voice queries (e.g. "Which restaurant? **A** or **B**?").
- Prototyped voice UIs for embedded devices (speakers, appliances).

Co-Founder & CEO — 2014 — 2016
Blue Tatami — *Cambridge, UK*

Co-founded Blue Tatami, an iOS personal productivity startup.
- Designed and shipped Dodeka, a 12-hour watch-face calendar app, from scratch.
- Grew to **6,000+** users with no paid advertising.



Projects

2017: Simple Datagram Protocol | Article – Bypassed MFi restrictions by using BTLE to configure Raspberry Pis from iOS.

2015: BioSwift – First open-source framework in Swift to handle gene data.

Presentations

2023 EuroLLVM - Prototyping MLIR in Python

2023 LLVM Dev Mtg - Compiler backend design with MLIR

2024 EuroLLVM - Teaching MLIR concepts to undergraduate students

2024 LLVM Dev Mtg - Quidditch: An End-to-End Deep Learning Compiler for Occamy using IREE & xDSL

Teaching

Taught introductory MLIR sessions at the 2025 MLIR Winter School, 2025 MLIR Summer School.

Skills

- **Programming Languages**: Python, Swift, Objective-C, C/C++, JavaScript/TypeScript, Assembly
- **Languages**: Fluent in English, Russian, French